



Problem statement:- Write a program to compute
i) Square root of number ii) Square of
number iii) Cube of number iv) Check for
prime v) Factorial of number vi) Prime factors
of given number using switch.

• Theory:-

(Switch statement.

The a switch statement in C is an alternate
to if-else if ladder statement which allows us to
execute multiple operations for the different
possible values of a single variable
called switch variable. Here, we can define
various statements in the multiple cases for
the different values of 'a single
variable'

The syntax of switch statement is in C
language is given below,

switch expression {

case value1;

//code to be executed

break; //optional

case value2;

//code to be executed

break; //optional

.....

default: code to be executed if all cases are not matched



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Rules for switch statement in C language

- 1) The switch expression must be of an integer or character type.
 - 2) The case value must be an integer or character constant.
 - 3) The case value can be used only inside the switch statement.
 - 4) The break statement in switch case is not mandatory. It is optional. If there is no break statement found in the case, all the types will be executed present after the matched case. It is known as fall through the state of a switch statement.
- Conclusion:- We have successfully written the menu driven program to compute square root, square, cube, check for prime, factorial, prime factor for a user entered number using a switch statement.


```

#include <stdio.h>
#include <math.h>

int main()
{
    int choice, n, i;
    int fact=1, prime=1;

    printf("\n 1. Square Root ");
    printf("\n 2. Square ");
    printf("\n 3. Cube ");
    printf("\n 4. Check Prime ");
    printf("\n 5. Factorial ");
    printf("\n 6. Prime Factors ");
    printf("\n Enter choice: ");
    scanf("%d", &choice);

    printf("\n Enter number: ");
    scanf("%d", &n);

    switch (choice)
    {
        case 1: printf("Square root = %.2f", sqrt(n));
                break;
        case 2: printf("Square = %d", n*n);
                break;
        case 3: printf("Cube = %d", n*n*n);
                break;
        case 4: if (n <= 1) prime = 0;
                for (i=2; i <= n; i++)
                {
                    if (n%i == 0)
                    {
                        prime = 0;
                        break;
                    }
                }
    }
}

```

```

if (prime == 1)
    printf("Prime number");
else
    printf("Not a Prime number");
case 5: for (i=1; i <= n; i++)
        fact = fact*i;
        printf("Factorial = %d", fact);
        break;
case 6: printf("Prime factors");
        for (i=2; i <= n; i++)
        {
            while (n%i == 0)
            {
                printf("%d ", i);
            }
        }
    }
}

```

```
if (prime == 1)
    printf("Prime number");
else
    printf("Not a Prime number");
Case 5 :- for (i = 1; i <= n; i++)
    fact = fact * i;
printf("Factorial = %d", fact);
break;
Case 6 :- printf("Prime factors");
for (i = 2; i <= n; i++)
{
    while (n % i == 0)
    {
        printf("%d ", i);
    }
}
```

```
} n = n/i;
```

```
}  
break;
```

```
default: printf("wrong choice");  
}  
return 0;
```